

Design Of A Modular Architecture Using ROS :

The Pepper Case

RoboBreizh Team

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- Downloaded from <http://ajph.org/> on November 10, 2014

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Figure 1: Our team with Pepper and Nao robots

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Architecture design

Examples of required capabilities for @HomeEDU

- Recognize human
- Recognize objects
- Navigate safely
- Understand basics commands
- Take tiny objects
- Bonus : act natural

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Architecture design

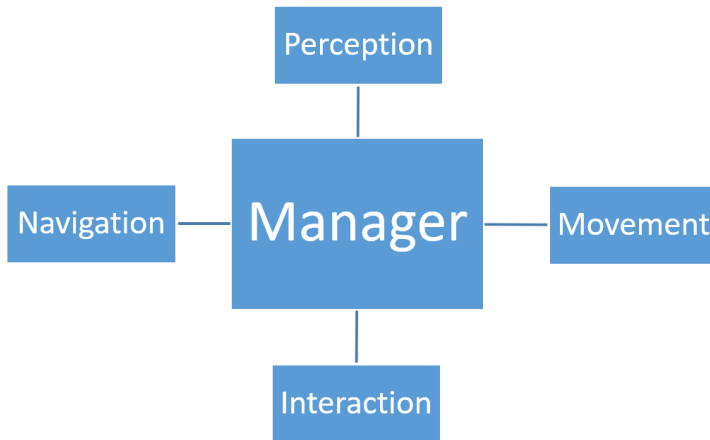


Figure 3: **First step:** split into modules

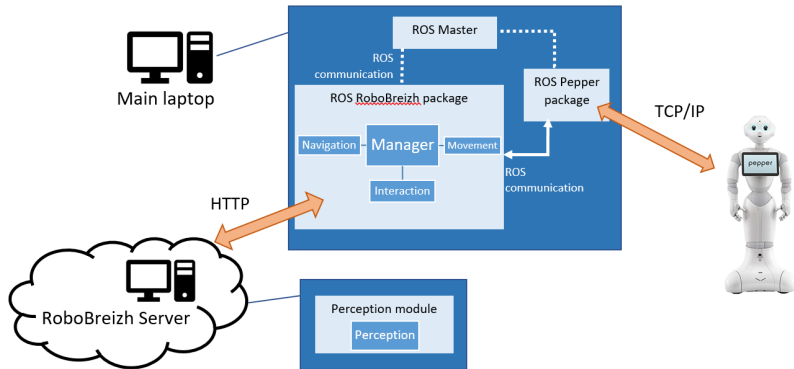


Figure 5: **Third step:** wrap it up

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ROS Basics

How to control every robot the same way?



ROS: Robot Operating System

- Created by Willow Garage in 2007
- An open-ended collaboration framework
- Over 150 compatible robots
- Over 3,000 available packages
- A large community

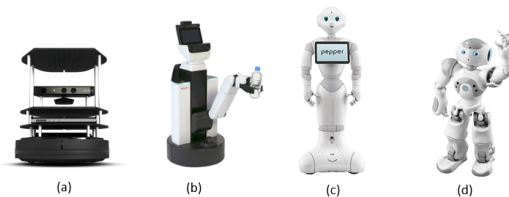


Figure 7: Example of compatible robots: (a) TurtleBot, (b) HSR (c) Pepper, (d) Nao

Architecture of ROS communication:

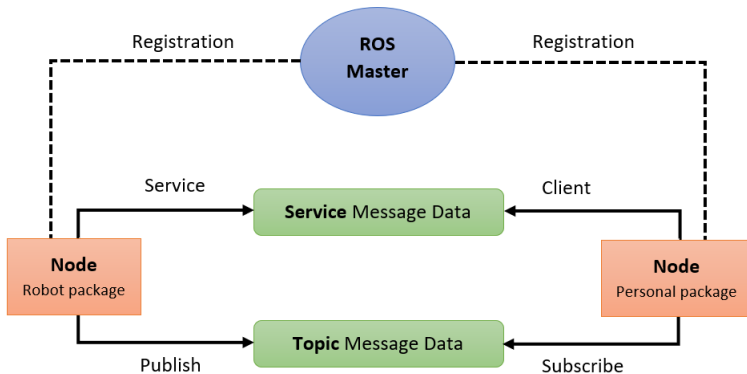


Figure 8: Communication infrastructure

Lots of tools:

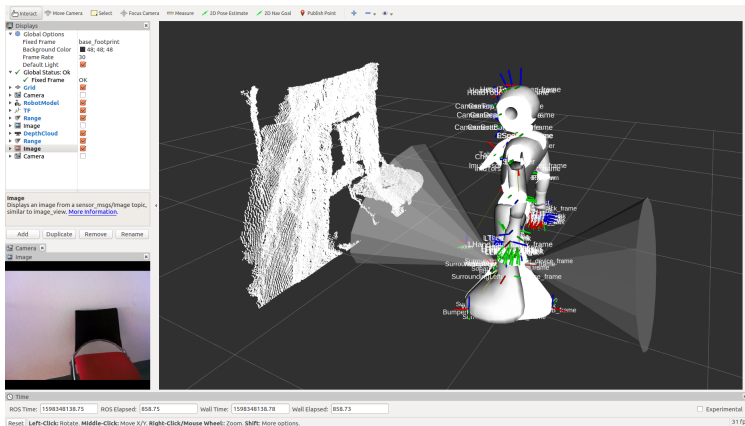


Figure 9: **RViz**: 3D visualization of many sensors

ROS Basics

Discussion

- ✓ Open-source
- ✓ Standard framework for multiple robots
- ✓ A large community
- ✓ Lots of available packages and tutorials
(<http://wiki.ros.org/ROS/Tutorials>)
- ✗ May not have all possibilities given by the native API of the robot
- ✗ Hard to understand for beginner in computer science
- ✗ Asynchronous communication: not adapted to a sequential scenario

The Manager

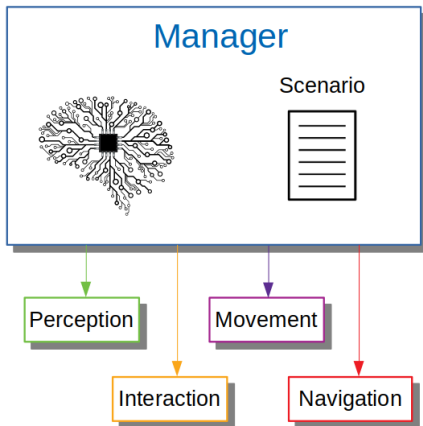


Figure 10: Manager's aim and links

The Manager

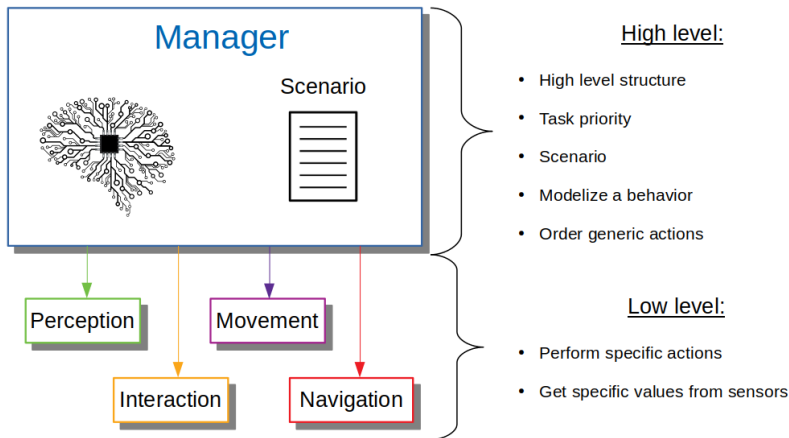


Figure 11: Manager's aim and links

The Manager: Discussion

Discussion

- ✓ Answer to our needs
- ✓ Links modules together
- ✓ High level structure
- ✓ Simple to make and use
- ✓ Time interval limit for order's execution
- ✗ Not a state machine
- ✗ Actions are performed sequentially
- ✗ Scenario paused during order's execution
- ✗ Not fully developed, can be enhance

Perception Module

What is computer vision?

- Aim to reproduce the human's sight
- Deduce whether objects are present in a picture
- Deduce where they are located

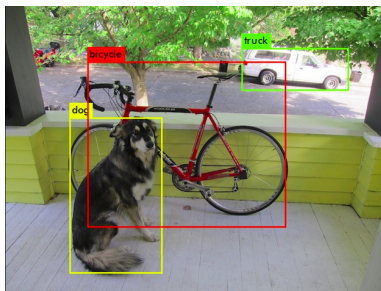


Figure 12: Object detection using YOLOv3 [13]

Perception Module

@HomeEDU requirements

- Detect objects (e.g. bags, chairs...)
- Detect persons
- Bonus: Detect a person's movement

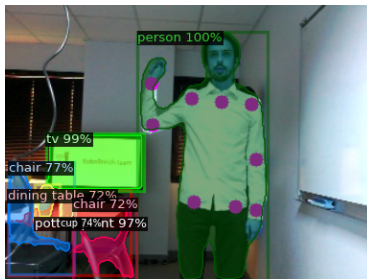
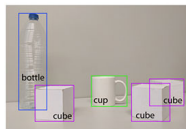


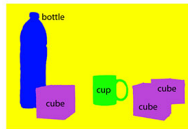
Figure 13: What and how much can be detected?



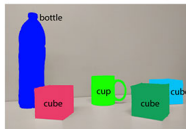
(a) Image classification



(b) Object localization



(c) Semantic segmentation



(d) Instance segmentation

Figure 14: Detecting objects using Mask R-CNN [9]

Perception Module

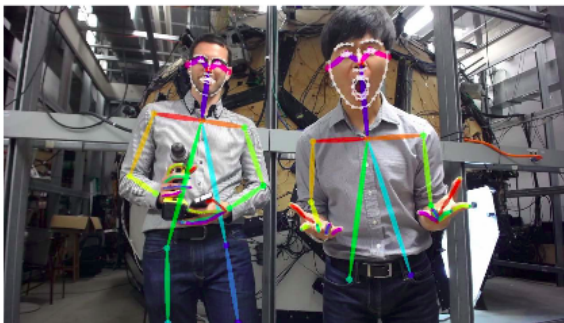


Figure 15: Pose estimation: OpenPose [4]

Perception Module

Constraints

- Require computer intensive operations

What can be performed in real-time?

- Pepper's camera has a poor resolution.

Is the image quality good enough to extract the information we need?

Perception Module

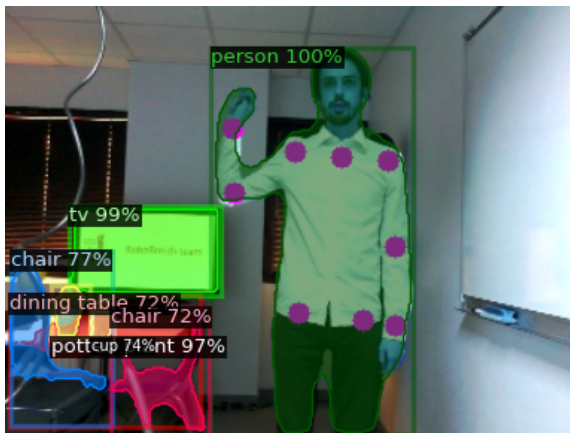


Figure 16: Mask-RCNN and OPenPose video demonstration

Perception Module

Discussion

- ✓ High performance in real-time
- ✓ Can detect on a pixel level
- ✓ Can detect poses
- ✗ Requires a computer with a GPU
- ✗ Communication between the robot and the computer adds latency

1 Architecture design

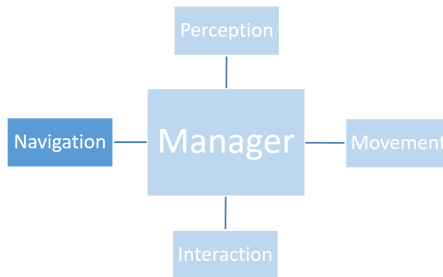
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Navigation Module

Navigation in Robotics:

- Environment representation (i.e. Mapping)
- Real Time Localization
- Pathfinding

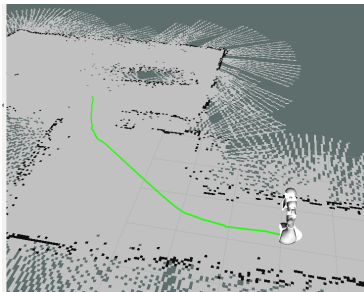


Figure 17: Navigation example

Navigation Module

@HomeEDU Requirements

Safely navigate to a given 2D coordinates

Others

- Navigate in known environment
- Easy-to-use and fast implementation
- Deal with known obstacles

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Navigation Module

Constraints

- Accuracy of Pepper depth camera
- Pepper number and range of lasers

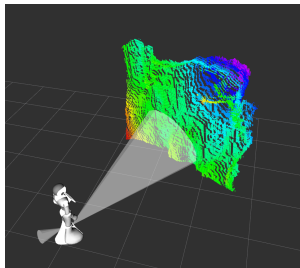


Figure 20: Pepper looking at a flat wall
- Rviz Visualization of depth camera

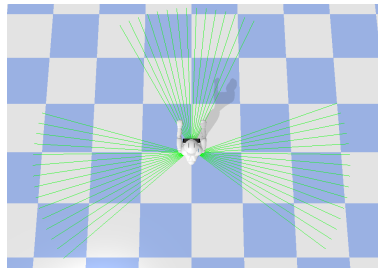


Figure 21: Pepper lasers' range and position

Navigation Module

Our approach with Pepper

- LiDAR based SLAM instead of Visual SLAM
- Mapping: Gmapping ROS node [7]
- Navigation: ROS Navigation Stack using AMCL (Adaptive Monte-Carlo Localization) [6] & move_base [5]

Navigation Module

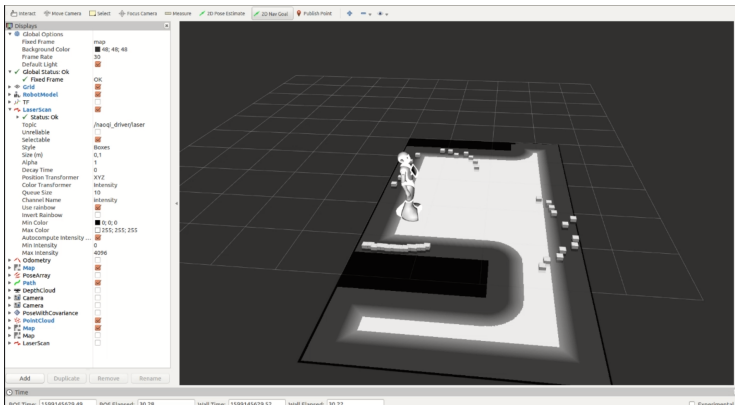


Figure 22: Map and pathfinding visualization through Rviz

Navigation Module



Figure 23: Autonomous Navigation in known environment

Navigation Module

Discussion

- ✓ Easy-to-use and fast implementation
- ✓ Work well with short path
- ✗ Inaccuracy of depth camera
- ✗ Non-map objects and moving people led sometimes the planner to failure

1 Architecture design

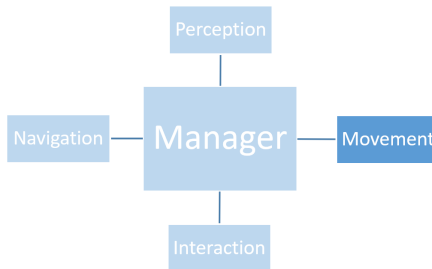
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Movement Module

What is a robotics movement?

- Allow the robot to move its arms and head
- Create a natural interaction with a human
- Interact with the environment

@HomeEDU requirements

- Specific movements (e.g. take objects, point a seat...)

Movement Module

State-of-the-art

- Used tools: naoqi_pose package (ROS)
- Other tools: NAOqi Motion, MoveIt! [12]
- Used learning algorithm: GMM (Gaussian Mixture Model) and GMR (Gaussian Mixture Regression) [3]

Constraints

- Only take light objects with Pepper
- Limited movements due to robot's reachable workspace
- Easy learning a movement

Movement Module

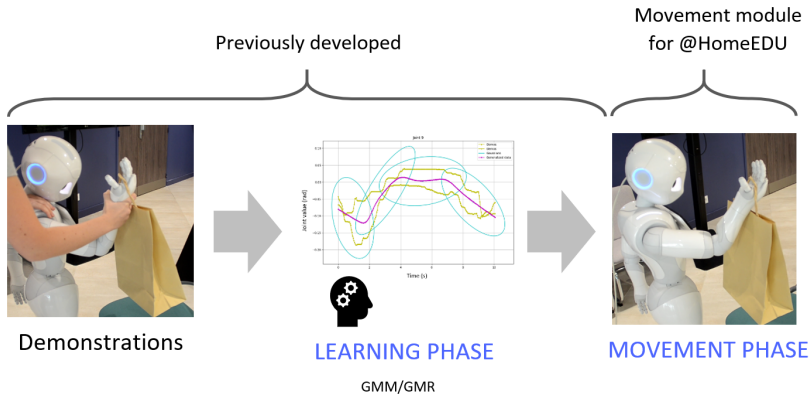


Figure 24: Learning movements process

Movement Module



Figure 25: Movement demonstration



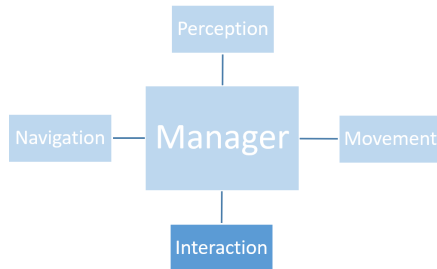
Figure 26: Learned movement

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Interaction Module

What is an interaction?

- A Human-Robot Interaction (HRI): a verbal and non-verbal communication
- **Verbal communication** : dialog
- **Non-verbal communication** : tablet, LEDs, gestures

@HomeEDU requirements

- Understand human speech and his intention
- Answer to a human and memorize information
- Natural interaction with non-verbal communication

Non-verbal communication

- Display information (tablet)
- Give feedback about its state (tablet, LEDs)
- Show its emotion (tablet, LEDs)
- Communication gestures (arm, head, body)



Figure 28: Pepper's tablet and LEDs

RoboCup@Home Education Online Challenge: Final

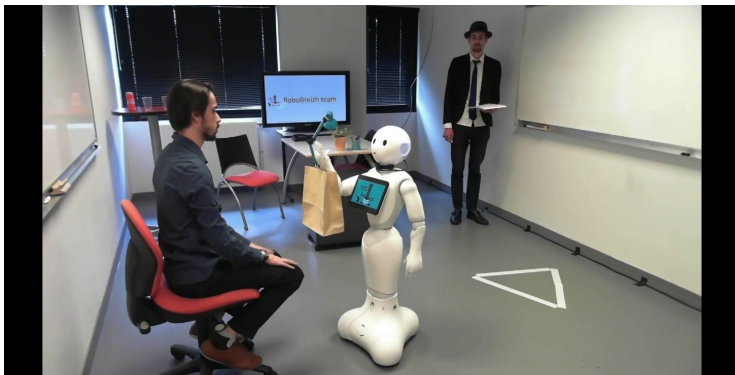


Figure 29: @HomeEDU final of RoboBreizh team

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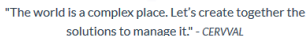
Summing up

- Choose the application and define the needs
- Define the module for the architecture and the used tools to create it (e.g. ROS)
- Be flexible
- RoboBreizh team won the Best Performance Award in Open Platform @HomeEDU

Conclusion

Futur work

- Improve the Dialog part
- Be on-board
- Improve Navigation system
- Participate in the RoboCup@Home competition



French National Research Agency



French National Centre for Scientific Research

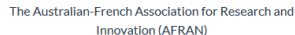


Figure 30: Our partners

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You only look once: Unified, real-time object detection.